AI-Based Ayurveda and Spiritual Wellness Tourism Matching System for Sri Lanka

Wellness Component of the Tour Ceylon Platform

Project ID – R26-IT-152

Project Proposal Report

De Silva D.C.M. — IT22108586

Bachelor of Science (Hons) Degree in Information Technology Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology, Sri Lanka (SLIIT)

October 2026

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
De Silva D.C.M.	IT22108586	

The above candidate is carrying out research for the undergraduate dissertation under my supervision.

Signature of the supervisor: _____ Date: _____

Table of Contents

List of Figures....4
List of Tables....4
ABSTRACT....5
Acknowledgement....6
LIST OF ABBREVIATIONS....7
LIST OF APPENDICES....8
1 INTRODUCTION....9
1.1 Background and Context....9
1.2 Literature Review....11
1.3 Research Gap....13
2 OBJECTIVES....14
2.1 Main Objective....14
2.2 Specific Objectives....14
3 METHODOLOGY....15
4 HIGH LEVEL ARCHITECTURE....18
5 USER REQUIREMENTS....19
6 COMMERCIALIZATION PLANS....21
7 GANTT CHART....23
8 SYSTEM DIAGRAM....24
9 CONCLUSION....25
10 BUDGET AND BUDGET JUSTIFICATION....26
11 WORK BREAKDOWN STRUCTURE....27
12 REFERENCES....28
13 APPENDICES....30
Appendix A: Dosha Lifestyle Questionnaire....30
Appendix B: Sample Centre Record....31
Appendix C: Preliminary Web Interface....32
Appendix D: Risk Analysis Table....34
Appendix E: Ethical Considerations....35

List of Figures

Figure 1: Three-Layer System Architecture Diagram.....18
Figure 2: Gantt Chart — March to November 2026 Project Timeline.....23
Figure 3: System Flowchart — End-to-End Recommendation Process.....24
Figure 4: Landing Page with the Ayurveda and Spiritual Paths.....32
Figure 5: Plain-Language Lifestyle Questions.....32
Figure 6: Ranked Results with Per-Factor Score Breakdown.....33

List of Tables

Table 1: Comparison of Existing Systems and Their Limitations.....9
Table 2: Summary of Literature Review.....12
Table 3: Technology Selection and Justification.....17
Table 4: Functional Requirements.....19
Table 5: Non-Functional Requirements.....19
Table 6: Budget Breakdown and Justification.....26
Table 7: Work Breakdown Structure.....27
Table 8: Dosha Lifestyle Questions.....30
Table 9: Fields of a Centre Record.....31
Table 10: Risk Analysis.....34

ABSTRACT

Sri Lanka recorded 2,362,521 international tourist arrivals in 2025, and 12,977 of these visitors stated health or Ayurveda as their main purpose of travel, staying an average of 14.7 days – longer than any other group [1]. Despite this demand, a foreign visitor who wants Ayurvedic treatment or a meditation retreat has no reliable way to find a centre that suits their health condition, Ayurvedic constitution (dosha), budget and language, and no structured way to judge the quality and safety of a centre before travelling. General platforms such as map services, review sites and retreat booking portals list centres by location and rating only.

This research proposes the Wellness Component of the Tour Ceylon platform: an explainable, multilingual AI system that matches international wellness tourists to suitable Ayurveda treatment centres and spiritual retreats in Sri Lanka. A Random Forest classifier will predict the tourist's dominant dosha from a short six-question lifestyle profile, selected after comparing six machine learning algorithms. A transparent weighted model will score a curated database of real Sri Lankan centres on four factors – condition match, dosha compatibility, review-derived quality and budget fit – and explain every score. Per-language TF-IDF and Support Vector Machine classifiers will interpret free-text symptom descriptions in English, German, Russian, Chinese and Japanese, and a hybrid keyword and machine learning layer will screen guest reviews for safety concerns. The system will also estimate treatment duration, likely outcome and trip cost.

Expected outcomes include a dosha classifier with at least 80% accuracy, a matching model that outperforms condition-only matching in an ablation study, recommendation responses under 200 ms and a full-stack web application integrated with the group platform. The work aligns with SDG 3, SDG 8 and SDG 12, and has a commercial pathway through a freemium tourist service, verified centre listings and a B2B matching API.

Keywords: Wellness Tourism, Ayurveda, Dosha Classification, Random Forest, Recommender System, Multilingual Text Classification, Sri Lanka

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Mr Buddika Harshnath, and co-supervisor, Prof. Samantha Rajapaksa, for their guidance in shaping this research. I also thank the project's domain expert, an Ayurvedic practitioner, for sharing knowledge of Ayurvedic constitution, treatment patterns and anonymised case information, and the members of group R26-IT-152 for their collaboration on the shared Tour Ceylon platform.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
DSR	Design Science Research
FR	Functional Requirement
GWI	Global Wellness Institute
LOO	Leave-One-Out (cross-validation)
MAE	Mean Absolute Error
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
PDPA	Personal Data Protection Act (Sri Lanka)
REST	Representational State Transfer
RF	Random Forest
SDG	Sustainable Development Goal
SLTDA	Sri Lanka Tourism Development Authority
SUS	System Usability Scale
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
WBS	Work Breakdown Structure

LIST OF APPENDICES

Appendix A: Dosha Lifestyle Questionnaire

Appendix B: Sample Centre Record

Appendix C: Preliminary Web Interface

Appendix D: Risk Analysis Table

Appendix E: Ethical Considerations

1 INTRODUCTION

1.1 Background and Context

Tourism is one of Sri Lanka's largest sources of foreign exchange. The Sri Lanka Tourism Development Authority (SLTDA) recorded 2,362,521 international arrivals in 2025, a 15.1% increase over 2024 [1]. Within this market, wellness tourism is a fast-growing, high-value segment. The Global Wellness Institute valued global wellness tourism at USD 830.2 billion in 2023 and projects USD 1,351.0 billion by 2028 [2]. Sri Lanka's traditional Ayurveda and its Buddhist meditation heritage give the country a distinctive position in this market: in 2025, 12,977 visitors came mainly for health or Ayurveda, Germany was the largest source market for this purpose (6,363 visitors), and these visitors stayed an average of 14.7 days [1].

Choosing an Ayurveda centre or meditation retreat is a high-stakes decision for a foreign visitor. Treatment programmes run for two to five weeks, the quality of centres varies widely, and Ayurveda personalises treatment according to a person's constitution (dosha) [3], [4]. Fonseka and Pavithra found that Ayurveda and holistic treatments are the most influential reason wellness tourists choose Sri Lanka, but that the sector lacks standardisation and certification [5]. Practitioners themselves report weak regulation, inconsistent quality and poor information sharing [6].

Today, a visitor must piece together information from map services, review sites, retreat portals and centre websites. None of these sources considers the visitor's condition or constitution, none summarises safety concerns from guest reviews, and few are available in the visitor's own language. This absence of personalised, trustworthy and multilingual guidance is the core problem this research addresses.

Analysis of Existing Systems and Their Limitations

The platforms currently available to wellness tourists in Sri Lanka lack personalisation and quality assurance, as Table 1 shows.

Table 1: Comparison of Existing Systems and Their Limitations

Platform	Core Capability	Critical Limitation for Wellness Tourists
Google Maps and review sites	Location, star rating and free-text reviews of most centres	No condition or constitution matching; quality and safety must be read review by review; no cost estimate

Platform	Core Capability	Critical Limitation for Wellness Tourists
BookRetreats.com [7]	Bookable Ayurveda and meditation retreats in Sri Lanka	Mostly premium listed retreats only; no dosha profiling; no safety screening
Online dosha quizzes (e.g. Banyan Botanicals) [8]	Self-assessment of dosha	Result is not linked to any provider, condition or budget, and is not designed for Sri Lanka
Centre websites and tour operators	Programme details from a single provider	No comparison across providers; quality varies; often English only
SLTDA information services	Registration information and statistics	No personalised recommendation or quality assessment

None of these platforms connects a visitor's constitution to suitable providers, summarises quality and safety from reviews, or serves the main source markets in their own languages. These gaps form the focus of this research.

Proposed Solution

This research proposes the **Wellness Component** – an AI-based Ayurveda and spiritual retreat matching system. A tourist chooses between an Ayurveda treatment path and a meditation or spiritual retreat path, states a condition or goal (optionally as free text or speech in five languages), answers four plain-language lifestyle questions and selects a budget. The system predicts the tourist's dosha, ranks real centres by condition, dosha, quality and budget, explains each score, flags safety-relevant reviews, and estimates the treatment duration, likely outcome and cost. This component is one of four within the group's Tour Ceylon platform (R26-IT-152), alongside the SOS Emergency Detection System (De Silva D.S.K), the AI Itinerary Generator (Wanniarachchi P.M.R) and the Tourism Risk and Context Intelligence System (Abinaya R). All components share a unified backend and database and communicate through REST APIs.

SDG Alignment

1. **SDG 3 – Good Health and Well-being:** helps visitors find appropriate and safe traditional health care, and surfaces safety concerns and unverified providers [9].
2. **SDG 8 – Decent Work and Economic Growth (Target 8.9):** directs high-value, long-stay wellness tourists to local Ayurveda centres, including small regional providers, and promotes Sri Lanka's traditional medicine heritage.

3. **SDG 12 – Responsible Consumption and Production (Target 12.b):** transparent quality and safety information supports responsible tourism choices and gives a structured view of the wellness tourism supply.

Research Cluster Alignment

This project is aligned with the AI Centre of Excellence (CEAI) at SLIIT, under the research cluster of Machine Learning and Data Science for Real World Decision Making.

1.2 Literature Review

The literature review covers four areas: wellness and Ayurveda tourism in Sri Lanka, computational assessment of Ayurvedic constitution, tourism and health-tourism recommender systems, and the machine learning techniques suitable for this problem. Table 2 summarises the key sources.

Wellness and Ayurveda tourism. Smith and Puczkó distinguish medical tourism, which treats illness, from wellness tourism, which maintains wellbeing, and place Ayurveda in the overlap between the two [10]. In Sri Lanka, Fonseka and Pavithra surveyed 50 wellness tourists and found that 34% visited specifically for Ayurveda and holistic healing; the range of Ayurveda treatments explained 42.6% of variance in destination choice, while the lack of standardisation and certification was a key weakness [5]. Dharmapala and Gnanapala interviewed 21 Ayurvedic practitioners and identified weak government support, staff shortages, inconsistent regulation and poor information sharing as barriers [6]. The World Health Organization encourages the safe integration and regulation of traditional medicine [11].

Ayurvedic constitution. Research has begun to give dosha and prakriti an objective basis. Prasher *et al.* found gene-expression and biochemical differences between extreme constitutional types [12], and Govindaraj *et al.* reported genome-wide differences between prakriti groups [13]. Rotti *et al.* developed structured prakriti assessment instruments [3]. Bidve, Mishra and Annapurna classified prakriti types with Multinomial Naïve Bayes and achieved 0.90 accuracy using a questionnaire of 147 features [14]. These studies show that dosha can be assessed from observable traits, but they stop at a label and do not connect it to providers; long questionnaires are also impractical for tourists.

Recommender systems. Collaborative filtering needs interaction histories that a new platform does not have; knowledge-based recommenders instead match explicit requirements to item attributes and suit rare, high-value decisions [15], [16]. Borràs *et al.* surveyed tourism recommenders and found that they rank attractions and accommodation by preferences and context, without health inputs [17]. Panteli *et al.* proposed a health-tourism recommender that ranks providers against an ideal profile, but without constitution, review text or safety signals [18]. Explanations increase users' trust in recommendations [19].

Reviews and machine learning techniques. Hu, Pavlou and Zhang showed that average ratings can misrepresent true quality and that review content carries additional information [20]. Random Forests are robust on small, mixed categorical datasets and provide feature importance [21]; Support Vector Machines with TF-IDF features work well for short text classification [22], [23]; and character n-grams suit languages written without spaces, such as Chinese and Japanese [24].

Table 2: Summary of Literature Review

Author(s) and Year	Focus	Method / Finding	Limitation for this Problem
Fonseka & Pavithra, 2021 [5]	Sri Lanka wellness tourism demand	Survey; Ayurveda is the main driver of choice	No decision-support system
Dharmapala & Gnanapala, 2025 [6]	Ayurveda tourism challenges	Interviews; regulation and information gaps	No technical solution
Rotti et al., 2014 [3]	Prakriti assessment	Structured assessment instruments	Clinical setting; not for tourists
Bidve et al., 2023 [14]	ML prakriti classification	Naïve Bayes, 0.90 accuracy, 147 features	Long questionnaire; no provider matching
Panteli et al., 2021 [18]	Health-tourism recommender	Multi-criteria ranking with diversity	No constitution, reviews or safety
Borràs et al., 2014 [17]	Tourism recommender survey	Preference- and context-based ranking	No health or condition inputs
Hu et al., 2006 [20]	Online review quality	Ratings biased; content informative	Not applied to wellness providers
Tintarev & Masthoff, 2007 [19]	Explanations in recommenders	Explanations build trust	General, not domain-specific

1.3 Research Gap

A review of the research and existing products (Section 1.2) revealed four gaps that this component addresses.

Gap 1 – No constitution-aware provider matching. Tourism and health-tourism recommenders do not use the traveller's Ayurvedic constitution, and dosha classification research does not connect its result to real providers [17], [18], [14].

Gap 2 – Impractical constitution assessment. Existing ML approaches rely on long questionnaires [14]; no study tests whether a short, tourist-friendly profile can predict dosha with useful accuracy.

Gap 3 – No structured quality and safety signals. Guest reviews contain hygiene and safety information, but no wellness platform summarises it or flags concerns before a visitor commits to a multi-week stay [20].

Gap 4 – No multilingual access for the main source markets. Germany, Russia, China and Japan are major source markets, but existing tools offer little support in these languages for describing symptoms and goals [1].

Research Problem

How can an explainable, multilingual AI system match international wellness tourists to suitable and safe Ayurveda centres and spiritual retreats in Sri Lanka, using their health condition, Ayurvedic constitution, budget and review-derived quality signals?

2 OBJECTIVES

2.1 Main Objective

To design, implement and evaluate an explainable, multilingual AI system that matches international wellness tourists to suitable Ayurveda treatment centres and spiritual retreats in Sri Lanka, using their condition, Ayurvedic constitution, budget and review-derived quality and safety signals.

2.2 Specific Objectives

1. Build a curated, verified dataset of real Ayurveda and meditation centres in Sri Lanka, enriched with conditions treated, price tier, location and guest reviews.
2. Train and select a classifier that predicts a tourist's dominant dosha from a short lifestyle questionnaire with at least 80% accuracy, justified against alternative algorithms.
3. Design a transparent weighted matching model (condition, dosha, quality and budget) and measure each factor's contribution through an ablation study.
4. Provide multilingual access through per-language free-text symptom classifiers and a five-language interface for the main source markets.
5. Screen guest reviews for safety concerns with a hybrid keyword and machine learning approach, and estimate treatment duration, outcome and cost.
6. Deliver the component as a tested, secure full-stack application integrated with the Tour Ceylon platform, and evaluate it for correctness, performance and usability.

3 METHODOLOGY

Research Approach

This research follows the Design Science Research (DSR) methodology, in which a technical artefact is designed and evaluated to solve a real-world problem. The DSR cycle proceeds through problem identification, objective definition, artefact design, artefact development, and evaluation. The artefact in this research is the wellness matching component. Development will be iterative: each iteration adds a capability, runs automated tests and model evaluations, and is reviewed with the supervisor.

Data Collection

- **Centre database:** real Ayurveda and meditation centres collected from Google Places listings, enriched with conditions treated, price tier, coordinates and up to four guest reviews each, and manually checked and annotated with a verification note.
- **Tourist profiles:** anonymised real patient cases from the domain expert, extended with variations generated from the expert's dosha rules, recording six features (age, gender, sleep pattern, digestion and appetite, cold or heat sensitivity, emotional tendency) and the dominant dosha.
- **Labelled reviews:** real guest reviews labelled as concerning or not concerning for safety.
- **Free-text examples:** short symptom and retreat-goal descriptions in English, German, Russian, Chinese and Japanese, labelled with the condition or retreat type.
- **Outcome cases:** anonymised treatment outcomes (age, dosha, condition, duration, improvement) from the domain expert.

Technical Methods and Algorithms

Dosha classification. Six algorithms – Random Forest, Decision Tree, a neural network, Naïve Bayes, Support Vector Machine and k-nearest neighbours – will be compared on the same stratified 80/20 split. Random Forest is expected to perform best because it handles non-linear combinations of categorical traits on small datasets and provides feature importance [21]; the selected model will be confirmed with 5-fold cross-validation [25].

Matching model. A knowledge-based weighted model is proposed because the platform has no interaction history (the cold-start problem) and every recommendation must be explainable [16], [19]. For the Ayurveda path the score of a centre will be:

$$S = 0.35 \cdot Condition + 0.20 \cdot Dosha + 0.25 \cdot Quality + 0.20 \cdot Budget$$

For the spiritual path the dosha term is removed and its weight redistributed. A travelling companion's condition can be added as a combined score. Each factor's contribution is shown to the user. An ablation study will remove each factor in turn and compare the quality, budget alignment and dosha compatibility of the top recommendation over simulated tourist queries.

Multilingual text classification. One TF-IDF and linear SVM classifier will be trained per language and domain (five languages × symptom and retreat domains), using word n-grams for English, German and Russian and character n-grams for Chinese and Japanese [22], [24]. Accuracy will be estimated with leave-one-out cross-validation because each dataset is small.

Review safety screening. Keyword rules will be combined with a class-balanced TF-IDF and SVM classifier, so that the rules catch known terms and the model reduces false positives such as idiomatic praise.

Outcome and cost estimation. A Random Forest regressor will predict the expected improvement from age, dosha, condition and duration, and expert rules will estimate duration and trip cost by price tier and number of travellers.

Model Training and Validation

All models will be trained in a single reproducible notebook with a fixed random seed, which prints each model's parameters and computes every metric from the data. Evaluation metrics will include accuracy, precision, recall and F1 (weighted and per class) for classifiers, mean absolute error and R^2 for the outcome model, and average quality, budget alignment and dosha compatibility for the ablation study. The software will be verified with automated API tests covering functional, validation, security and admin scenarios, a latency benchmark, and task-based user acceptance testing with the System Usability Scale.

Tools and Platforms

Table 3: Technology Selection and Justification

Component	Selected Technology	Justification	Alternatives Rejected
ML development	Python, scikit-learn, pandas, NumPy	Mature Random Forest, SVM and TF-IDF implementations; small CPU models suit the dataset size	TensorFlow (too heavy for ~100-example datasets); R (weaker web integration)
Backend framework	Flask (Python)	Lightweight REST API in the same language as the models; matches the group's shared Flask backend	Django (more overhead); Node.js (separate ML service needed)
Frontend	HTML, CSS, JavaScript	Served by Flask on the same origin; no build step; five-language interface	React (build tooling not needed for a single module)
Database	PostgreSQL with JSON fallback	Relational store shared with the group platform; JSON fallback for offline demonstration	MongoDB (group database is relational)
Maps	Leaflet with OpenStreetMap	Free and key-less map of matched centres	Google Maps API (requires billing)
Voice input	Web Speech API	Built into Chromium browsers; supports all five languages	Paid speech APIs
Training environment	Jupyter / Google Colab	Reproducible, shareable training without local setup	Local scripts only
Testing and version control	pytest, Git and GitHub	Automated regression tests; branch per component	Manual testing only

4 HIGH LEVEL ARCHITECTURE

The component follows a three-tier architecture, shown in Figure 1. The **presentation layer** contains the tourist web application (three-step wizard, results, details, map, cost calculator and scenario comparison in five languages) and an admin dashboard. The **API layer** is a Flask REST service that validates input and protects admin functions. The **matching engine** contains the dosha classifier, the weighted matcher, the text and safety classifiers, the outcome predictor and the expert duration and cost rules. The **data layer** stores centres and reviews in PostgreSQL, with an automatic JSON fallback, together with the trained model files produced by the training notebook.

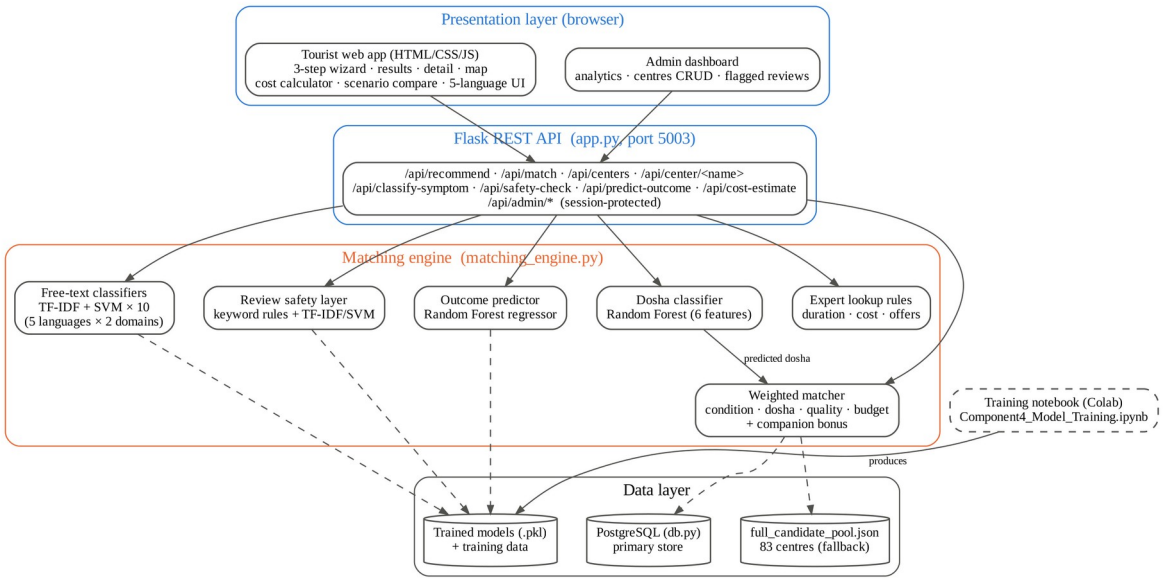


Figure 1: Three-Layer System Architecture Diagram

Within the Tour Ceylon platform, the wellness routes will be registered in the shared Flask backend under the same API paths, wellness tables will be created in the shared PostgreSQL database, and each matching session will be linked to the platform's tourist identifier so that a chosen centre and treatment duration can feed into itinerary planning.

5 USER REQUIREMENTS

Functional Requirements

Table 4: Functional Requirements

ID	Requirement
FR1	The system shall provide separate Ayurveda treatment and spiritual retreat paths with their own conditions and centres.
FR2	The system shall predict the tourist's dominant dosha and its confidence from age, gender and four lifestyle answers.
FR3	The system shall rank centres by condition, dosha, quality and budget and explain each score.
FR4	The system shall support a travelling companion with a second condition.
FR5	The system shall allow two scenarios (condition or budget) to be compared and show full centre details.
FR6	The system shall estimate treatment duration and trip cost for a number of travellers in several currencies.
FR7	The system shall suggest a condition or retreat type from free text or speech in five languages.
FR8	The system shall flag safety-relevant reviews and estimate a personalised treatment outcome.
FR9	The system shall provide an authenticated admin dashboard for analytics, centre management and review moderation.
FR10	The system shall show matched centres on a map with practical guidance such as a before-you-go checklist and seasonal advice.

Non-Functional Requirements

Table 5: Non-Functional Requirements

ID	Category	Requirement
NFR1	Performance	A recommendation shall be returned in under 200 ms (median) on a laptop-class machine.
NFR2	Reliability	Invalid input shall return a clear error message and never cause a server failure.
NFR3	Security	Admin functions shall require authentication; secrets shall be kept in environment variables.
NFR4	Usability	Tourists without Ayurvedic knowledge shall be able to complete the three-step wizard in their own language.
NFR5	Transparency	Each recommendation shall show its factor breakdown, and each prediction its confidence or model accuracy.
NFR6	Portability	The system shall run with PostgreSQL or, if unavailable, from a local data file.

ID	Category	Requirement
NFR7	Privacy	No personal identifiers shall be required, in line with Sri Lanka's PDPA [26].
NFR8	Reproducibility	All models shall be retrainable from one notebook with fixed seeds and pinned library versions.

User Roles

- **Tourist:** completes the wizard, views explained matches, compares scenarios, estimates cost and shares results.
- **Administrator:** maintains centre data, reviews flagged reviews and monitors analytics and model results.
- **Platform components:** consume the wellness API through the shared backend.

6 COMMERCIALIZATION PLANS

Target Market and Customer Persona

The primary B2C segment is international wellness tourists. SLTDA recorded 12,977 visitors whose main purpose was health or Ayurveda in 2025, with an average stay of 14.7 days; Germany alone contributed 6,363 of them [1]. A typical persona is a European visitor in their 40s or 50s with a chronic condition such as joint pain or stress, planning a two- to three-week Ayurveda stay, who wants a trustworthy centre within a mid-range budget and prefers information in their own language.

The secondary B2B segment includes Ayurveda centres and meditation retreats that want to reach qualified foreign guests, and tour operators and travel platforms that package wellness stays.

Value Proposition

For tourists, the system turns an opaque, high-stakes choice into a guided decision: personalised and explained matches, safety flags and a clear cost estimate in their own language. For centres, it rewards good service through review-derived quality and gives smaller, well-reviewed centres visibility alongside large brands. The global wellness tourism market, projected to reach USD 1,351.0 billion by 2028 [2], indicates strong long-term demand.

Revenue Model

Phase 1 – Freemium tourist service: matching and explanations are free; premium features such as saved plans, detailed reports and enquiry handling may be offered per trip.

Phase 2 – Verified centre listings and booking commission: centres that pass verification (registration, qualified physician, hygiene) pay a monthly listing fee for a verified badge and richer profile, and a commission is charged on enquiries that convert into stays.

Phase 3 – B2B matching API: tour operators and travel platforms license the matching service to embed it in their own products.

Competitive Advantage

The system is the only option identified that connects a dosha assessment to real Sri Lankan providers, summarises quality and safety across reviews, and serves the main source markets in

their own languages. The curated and verified dataset and the explainable matching model are difficult to replicate, and integration with the Tour Ceylon platform links the wellness choice to itinerary planning and safety services.

Projected Revenue (Conceptual Estimate)

As a conservative estimate, if 5% of health and Ayurveda visitors (about 650 per year) book through the platform with an average booking value of LKR 300,000 and a 10% commission, commission revenue would be about LKR 19.5 million per year. Thirty verified centres paying LKR 6,000 per month would add about LKR 2.2 million per year. These figures are assumptions for illustration and must be validated with centres.

Intellectual Property Considerations

The intellectual assets of the project include the curated and annotated dataset of Sri Lankan wellness centres, the trained dosha and text classifiers, and the matching model with its explanation method. Third-party libraries are used under permissive open-source licences, and map data from OpenStreetMap requires attribution. A commercial version must comply with the PDPA for health-related data [26] and with the terms of any third-party review data source.

7 GANTT CHART

Figure 2 shows the project timeline from March to November 2026, including the assessment milestones.

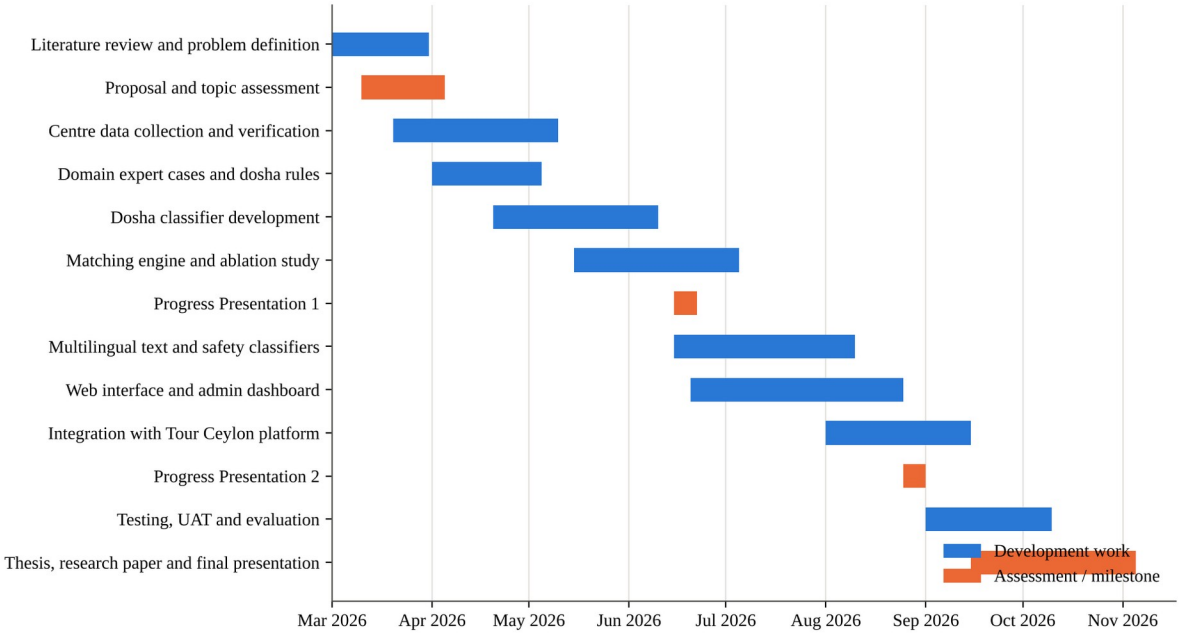


Figure 2: Gantt Chart — March to November 2026 Project Timeline

8 SYSTEM DIAGRAM

Figure 3 shows the end-to-end processing of a recommendation request, from the tourist's profile to the enriched, explained result list.

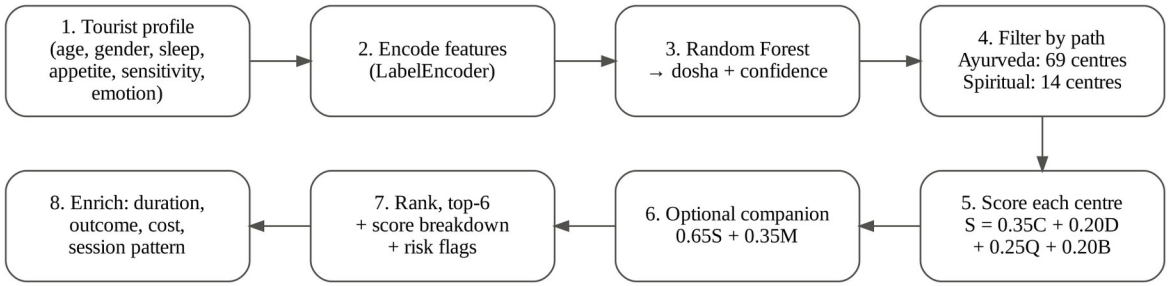


Figure 3: System Flowchart — End-to-End Recommendation Process

9 CONCLUSION

This proposal presents the Wellness Component of the Tour Ceylon platform, an explainable and multilingual AI system that matches international wellness tourists to suitable Ayurveda centres and spiritual retreats in Sri Lanka. By combining a short dosha profile classified with a Random Forest, a transparent four-factor matching model over a curated database of real centres, review-derived quality and safety signals, and per-language text classifiers, the system addresses four gaps identified in the literature: the absence of constitution-aware provider matching, the impracticality of long constitution questionnaires, the lack of structured quality and safety information, and the lack of multilingual access for the main source markets.

The methodology, grounded in Design Science Research and evaluated with cross-validation, an ablation study, automated testing and user acceptance testing, is feasible within the project timeline and on ordinary hardware. If the target of at least 80% dosha accuracy and an improvement over condition-only matching is achieved, the system will be a novel and deployable contribution to Sri Lanka's wellness tourism, aligned with SDG 3, SDG 8 and SDG 12 and with a credible pathway to commercialisation.

10 BUDGET AND BUDGET JUSTIFICATION

Table 6: Budget Breakdown and Justification

Item	Description	Cost (LKR)	Justification
Data collection and verification	Transport for visits to selected centres to verify listings and record information	15,000	Verification of centre data is essential for trustworthy recommendations
Domain expert consultation	Sessions with the Ayurvedic practitioner for case data and rule validation	5,000	Expert knowledge is needed for dosha rules, durations and outcomes
Cloud hosting	Small server instance for the Flask service and database during integration and testing (3 months)	9,000	Required for group integration testing and demonstration
User acceptance testing	Refreshments or tokens of appreciation for test participants	5,000	Encourages participation in task-based UAT
Development tools	Python, scikit-learn, Flask, Leaflet, Google Colab (free tier)	0	All tools are open source or free
Miscellaneous	Printing, binding and contingency	4,000	Final submissions and unexpected costs
Total		**38,000**	

All machine learning and web libraries are open source. Hosting costs are shared with the group where the infrastructure is shared.

11 WORK BREAKDOWN STRUCTURE

Table 7: Work Breakdown Structure

Task ID	Task	What Gets Delivered	When
T1.1	Review literature on wellness tourism, dosha assessment and recommenders	Literature review and research gap	March 2026
T1.2	Collect centre listings and guest reviews; verify and annotate centres	Curated centre database	March–May 2026
T1.3	Obtain anonymised expert cases and dosha rules; build profile dataset	Tourist profile dataset	April–May 2026
T2.1	Compare six algorithms and train the dosha classifier	Selected model and evaluation	April–June 2026
T2.2	Design and implement the weighted matching engine	Matching engine with score breakdown	May–June 2026
T2.3	Run the ablation study	Ablation results	June–July 2026
T3.1	Build multilingual symptom and retreat classifiers	Ten per-language classifiers	June–August 2026
T3.2	Build the review safety layer and outcome predictor	Safety and outcome models	July–August 2026
T4.1	Develop the Flask API and five-language web interface	Working web application	June–August 2026
T4.2	Develop the admin dashboard	Analytics and centre management	July–August 2026
T5.1	Integrate with the shared backend and database	Integrated component	August–September 2026
T5.2	Automated testing, benchmark and UAT	Test report and UAT results	September–October 2026
T6.1	Thesis, research paper and final presentation	Final deliverables	September–November 2026

12 REFERENCES

[1] Sri Lanka Tourism Development Authority, "Year in Review, 2025," SLTDA, Colombo, Sri Lanka, 2026. [Online]. Available: https://www.sltda.gov.lk/storage/common_media/Year_in_Review_2025_Final_updated_Report_2026_04_02.pdf

[2] Global Wellness Institute, "Global Wellness Economy Monitor 2024," Global Wellness Institute, Miami, FL, USA, Nov. 2024. [Online]. Available: https://globalwellnessinstitute.org/wp-content/uploads/2024/11/WellnessEconMonitor2024PDF.pdf

[3] H. Rotti *et al.*, "Determinants of prakriti, the human constitution types of Indian traditional medicine and its correlation with contemporary science," *Journal of Ayurveda and Integrative Medicine*, vol. 5, no. 3, pp. 167–175, 2014.

[4] S. Dey and P. Pahwa, "Prakriti and its associations with metabolism, chronic diseases, and genotypes: Possibilities of newborn screening and a lifetime of personalized prevention," *Journal of Ayurveda and Integrative Medicine*, vol. 5, no. 1, pp. 15–24, 2014.

[5] N. Fonseka and K. H. Pavithra, "Analysis of the potentials for promoting Sri Lanka as a wellness tourism destination," *SEUSL Journal of Marketing*, vol. 6, no. 1, pp. 44–52, 2021.

[6] M. V. S. E. Dharmapala and W. K. A. C. Gnanapala, "Issues and challenges related to Ayurveda wellness tourism development in Sri Lanka: Perspective of Ayurveda health professionals," in *Proc. Annual Research Symposium 2025*, University of Colombo, Sri Lanka, 2025, p. 682.

[7] BookRetreats.com, "Ayurvedic retreats in Sri Lanka," 2026. [Online]. Available: https://bookretreats.com/s/yoga-retreats/ayurvedic-retreats/sri-lanka

[8] Banyan Botanicals, "Dosha quiz," 2026. [Online]. Available: https://www.banyanbotanicals.com/pages/dosha-quiz

[9] United Nations General Assembly, "Transforming our world: The 2030 Agenda for Sustainable Development," Resolution A/RES/70/1, Oct. 2015.

[10] M. Smith and L. Puczkó, *Health, Tourism and Hospitality: Spas, Wellness and Medical Travel*, 2nd ed. Abingdon, U.K.: Routledge, 2014.

[11] World Health Organization, *WHO Traditional Medicine Strategy: 2014–2023*. Geneva, Switzerland: WHO, 2013.

[12] B. Prasher *et al.*, "Whole genome expression and biochemical correlates of extreme constitutional types defined in Ayurveda," *Journal of Translational Medicine*, vol. 6, Art. no. 48, 2008.

[13] P. Govindaraj *et al.*, "Genome-wide analysis correlates Ayurveda Prakriti," *Scientific Reports*, vol. 5, Art. no. 15786, 2015.

[14] P. Bidve, S. Mishra, and Annapurna J, "Enhancing Ayurvedic diagnosis using multinomial naïve Bayes and K-modes clustering: An investigation into prakriti types and dosha overlapping," arXiv:2310.02920, 2023.

[15] G. Adomavicius and A. Tuzhilin, "Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions," *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734–749, 2005.

[16] R. Burke, "Hybrid recommender systems: Survey and experiments," *User Modeling and User-Adapted Interaction*, vol. 12, no. 4, pp. 331–370, 2002.

[17] J. Borràs, A. Moreno, and A. Valls, "Intelligent tourism recommender systems: A survey," *Expert Systems with Applications*, vol. 41, no. 16, pp. 7370–7389, 2014.

[18] A. Panteli, A. Kompothrekas, C. Halkiopoulos, and B. Boutsinas, "An innovative recommender system for health tourism," in *Culture and Tourism in a Smart, Globalized, and Sustainable World*, V. Katsoni and C. van Zyl, Eds. Cham, Switzerland: Springer, 2021, pp. 649–658.

[19] N. Tintarev and J. Masthoff, "A survey of explanations in recommender systems," in *Proc. IEEE 23rd Int. Conf. Data Engineering Workshop*, Istanbul, Turkey, 2007, pp. 801–810.

[20] N. Hu, P. A. Pavlou, and J. Zhang, "Can online reviews reveal a product's true quality? Empirical findings and analytical modeling of online word-of-mouth communication," in *Proc. 7th ACM Conf. Electronic Commerce*, Ann Arbor, MI, USA, 2006, pp. 324–330.

[21] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[22] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.

[23] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Information Processing & Management*, vol. 24, no. 5, pp. 513–523, 1988.

[24] W. B. Cavnar and J. M. Trenkle, "N-gram-based text categorization," in *Proc. 3rd Annual Symp. Document Analysis and Information Retrieval (SDAIR-94)*, Las Vegas, NV, USA, 1994, pp. 161–175.

[25] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. 14th Int. Joint Conf. Artificial Intelligence (IJCAI)*, Montreal, QC, Canada, 1995, pp. 1137–1143.

[26] Parliament of the Democratic Socialist Republic of Sri Lanka, *Personal Data Protection Act, No. 9 of 2022*. Colombo, Sri Lanka: Government Publications Bureau, 2022.

13 APPENDICES

Appendix A: Dosha Lifestyle Questionnaire

The tourist answers four plain-language questions (in addition to age and gender). Each answer option corresponds to a typical Vata, Pitta or Kapha trait.

Table 8: Dosha Lifestyle Questions

Question	Typical Vata answer	Typical Pitta answer	Typical Kapha answer
How do you usually sleep?	Light sleeper, wakes up easily	Sleeps okay but gets warm at night	Deep, heavy sleeper
How is your appetite?	Varies a lot from day to day	Gets hungry or irritable if a meal is skipped	Steady; can skip meals easily
What is your temperature preference?	Feels the cold easily	Feels the heat easily	Not bothered much
Under stress, you tend to…	Worry or overthink	Get irritable or impatient	Stay calm; need a push to act

Appendix B: Sample Centre Record

Table 9: Fields of a Centre Record

Field	Example
Name	The Ayurveda National Hospital of Sri Lanka
Category	Curative
Conditions treated	General Ayurvedic medicine, teaching hospital
Dosha focus	General / Unspecified
Price tier	Budget (Government)
District	Kotte / Colombo
Verification note	Yes (Government)
Google rating and review count	4.3 (156 reviews)
Quality score and source	Derived from real guest reviews

Appendix C: Preliminary Web Interface

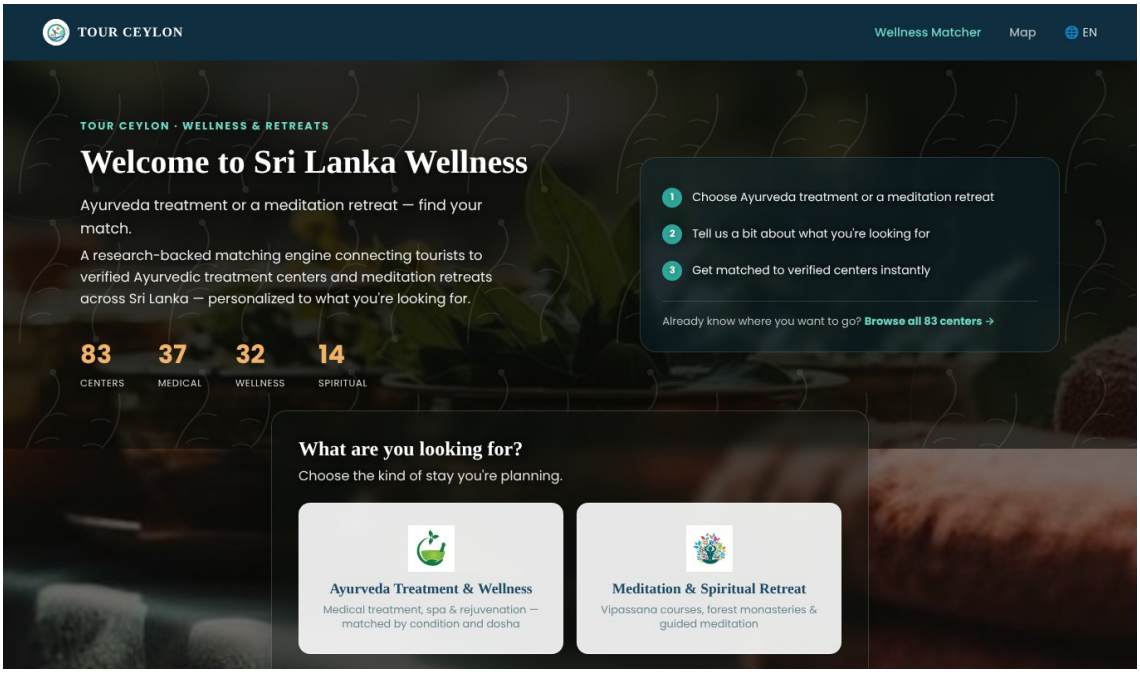


Figure 4: Landing Page with the Ayurveda and Spiritual Paths

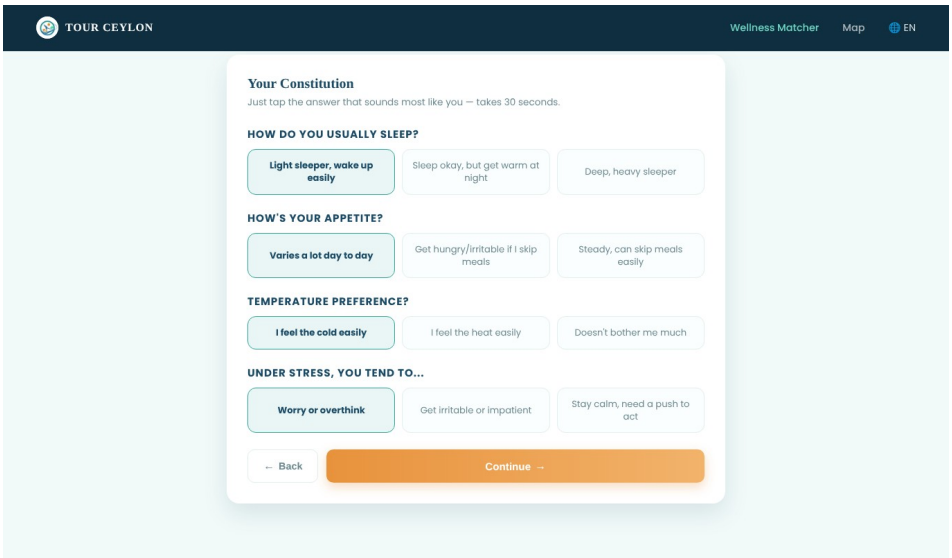


Figure 5: Plain-Language Lifestyle Questions

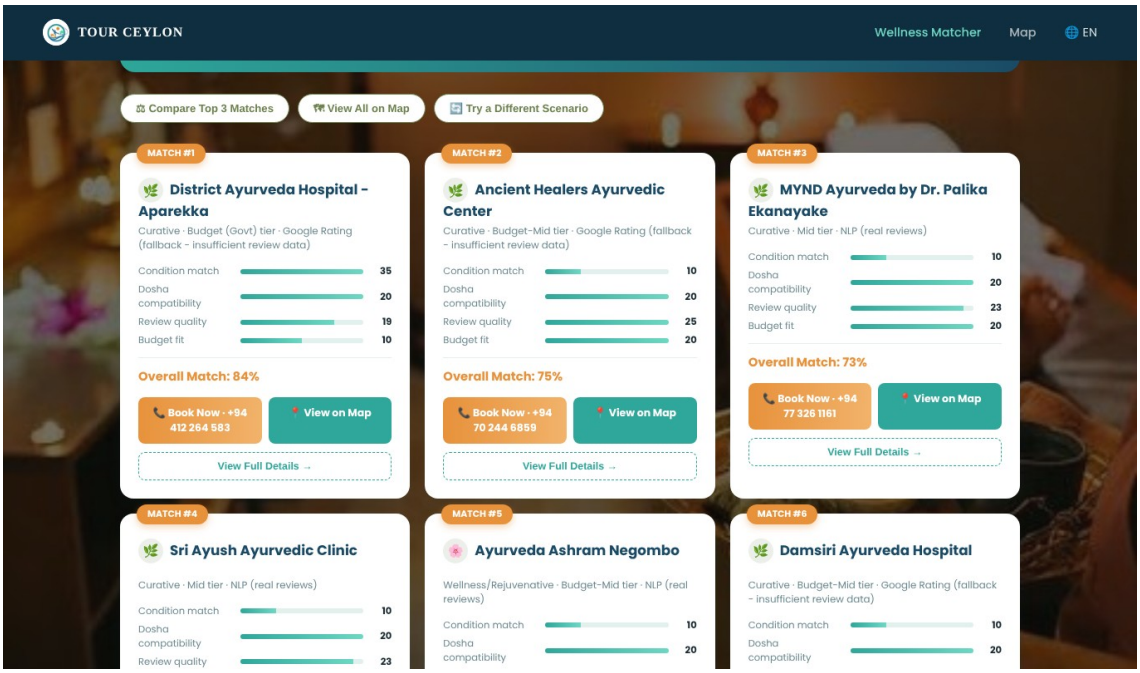


Figure 6: Ranked Results with Per-Factor Score Breakdown

Appendix D: Risk Analysis Table

Table 10: Risk Analysis

Risk	Likelihood	Impact	Mitigation
Too few real labelled profiles for the dosha model	High	High	Expert-rule variations of real cases; report the limitation; collect real assessments later
Centre information becomes outdated	Medium	Medium	Admin dashboard for updates; verification labels
Low accuracy for some languages	Medium	Medium	Report per-language accuracy; no-signal response instead of guessing; dropdown fallback
Few concerning reviews for the safety model	High	Medium	Hybrid keyword and ML design; manual moderation in the admin dashboard
Library version mismatch during integration	Medium	High	Pin library versions; automated tests in the shared environment
Low UAT participation	Medium	Medium	Recruit through tourist networks and the domain expert; small incentives

Appendix E: Ethical Considerations

- **Health-related data:** the system will not require names or contact details and will not store tourist queries in the standalone component, in line with the PDPA [26].
- **Anonymised expert data:** patient cases will be provided without names or identifiers.
- **Decision support, not diagnosis:** the interface will state that the system does not replace a consultation with a qualified Ayurvedic physician, and will show the confidence or accuracy of every prediction.
- **Informed consent:** UAT participants will receive an information sheet and give consent; no personal data will be published.
- **Fairness:** model performance will be reported per class and per language so that weaker groups are visible.